Value Stream prediction — representation EDA

Question: for VS prediction, what text representation works best in each of the three places text is used — and can we drop summarization to save cost?

Setup (held constant across every run): - 100 tickets with **≥3 ground-truth VS** (`--min-gt 3 --sample 100`, fixed seed) — the hard, discriminating cases; single-VS tickets wash out differences. - `--count-mode gt` — request exactly the GT count, so micro **F1 = P = R** (one clean number per run) and no count-generosity bias. - evidence mode, the Recall selection prompt, K=6 historic neighbours, judge on.

Terms & token counts (read this first)

Every IDMT ticket produces **two forms of text**. The whole study is about which form goes where.

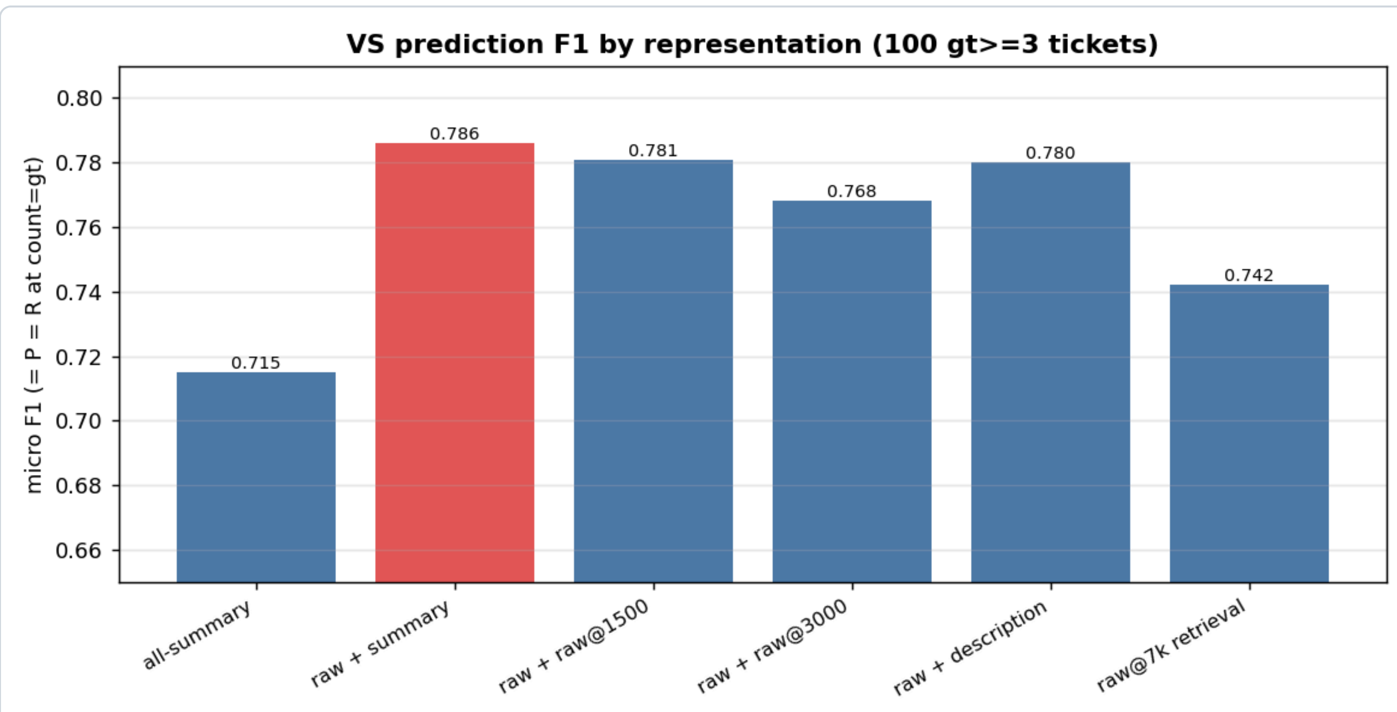
term	what it actually is	size
raw	the ticket's description **+** extracted attachment text, consolidated and capped during ingest	**≤ 24,000 tokens** (`doc_char_budget = 96k chars`)
summary	the condense LLM's output from that raw — 6 fields: generatedSummary, businessProblem, businessCapability, keyTerms, stakeholders, systemsAndProducts	**~460 tokens** (all 6 concatenated)
description	only the ticket's Jira description field (no attachments)	a few hundred tokens
snippet	the first ~200 characters of a stored doc	~50 tokens
`raw@Nk`	the **raw** truncated to N-thousand tokens (e.g. `raw@7k` = first ~7,000 tokens of the raw)	N,000 tokens

These forms get used in **three independent places** per prediction:

place	what it does	what goes here (winner)
Retrieval query	the text we **embed** to search the index for the 6 nearest past tickets	the **summary** (~460 tok)
New-ticket prompt	the new ticket's text the LLM **reads to pick the VS**	the **full raw** (~24,000 tok)
Historic block	each of the 6 retrieved neighbours, shown as precedent in the prompt	each neighbour's **summary** (~460 tok × 6) + its VS labels

So a run labelled **"raw + summary"** in the tables below means: **new-ticket prompt = raw**, **historic block = summary** (the retrieval query stays the summary unless the row says `raw@7k` retrieval). The ladder varies one place at a time.

Results



Columns = the form used in each of the three places (see the terms table above).

run	new-ticket prompt	historic block	retrieval query	F1	exact-set	avg latency
all-summary	summary	summary	summary	0.715	23%	4.2s

run	new-ticket prompt	historic block	retrieval query	F1	exact-set	avg latency
raw + summary	**raw**	**summary**	**summary**	**0.786**	**36%**	5.8s
raw + raw@1500	raw	raw@1500	summary	0.781	26%	5.6s
raw + raw@3000	raw	raw@3000	summary	0.768	24%	6.3s
raw + description	raw	description	summary	0.780	31%	4.4s
raw + raw@7k	raw	raw@7k	summary	0.780	30%	9.4s
raw@7k retrieval	raw@7k	raw@7k	**raw@7k**	**0.742**	23%	6.6s

The last row drops the summary entirely (raw-embedded index + raw everything). On a clean 100/100 with cheap raw@3k historic it lands at **0.742** — below the pack. (An earlier raw@7k-*historic* variant cost 13.9s avg / 132s max latency; dropping it to 3k fixed the latency but not the quality.)

How to read it: the big step is summary→raw on the **new-ticket prompt** (0.715 → 0.786). After that, swapping the **historic block** representation barely moves F1 (0.768–0.786) — and the last bar (raw@7k *retrieval*) drops *below* the pack. So the prompt is the lever, the historic block is a wash, and raw retrieval is a regression.

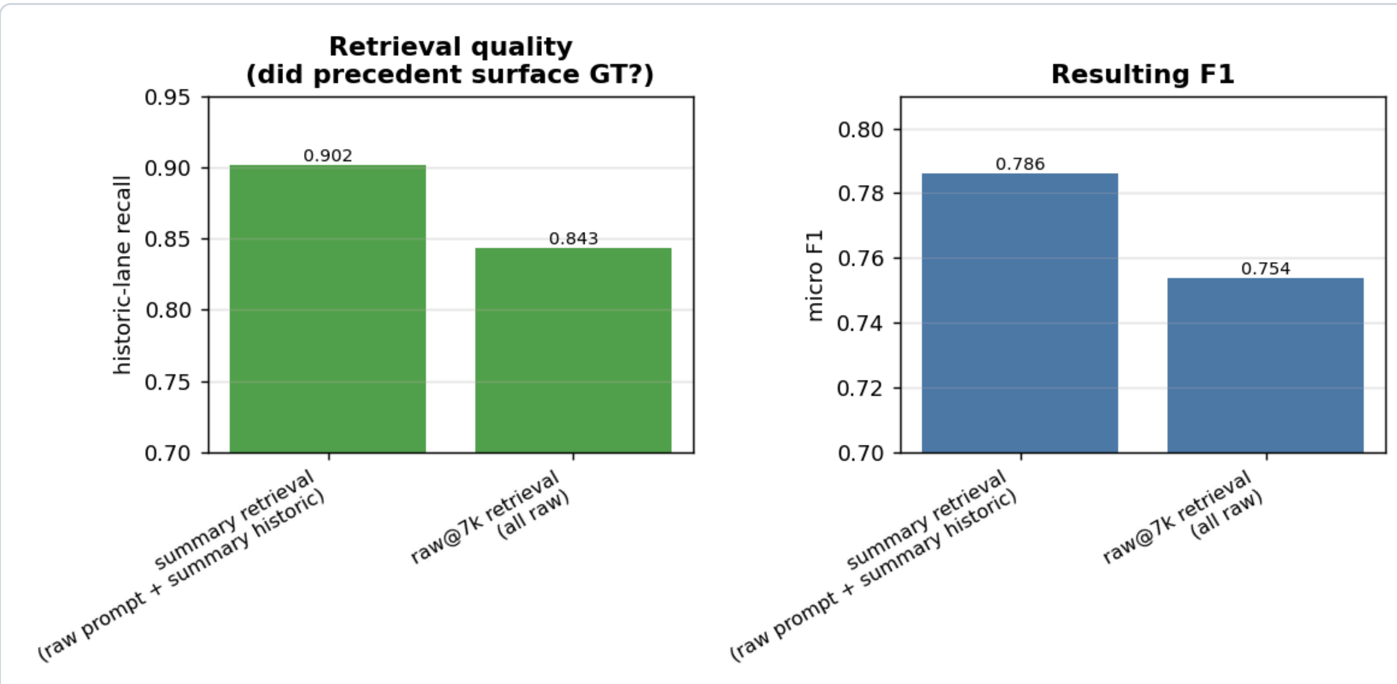
Finding 1 — the lever is the NEW-TICKET prompt

Feeding the new ticket's **raw text** instead of its summary is **+0.071 F1 (0.715 → 0.786)** and nearly doubles exact-set (23% → 36%). Retrieval is already perfect at surfacing candidates (every GT reaches the LLM), so all of this gain is the LLM *choosing* better when it sees the full ticket.

Finding 2 — the historic block representation is a wash (so use the cheapest)

Among the raw-prompt runs the historic block spans only 0.768–0.786: summary (0.786) ≈ description (0.780) ≈ raw@1500 (0.781) ≈ raw@7k (0.780), with raw@3000 slightly *worse* (0.768). More raw precedent does **not** help — it dilutes. **Summary historic is marginally best and the cheapest.**

Finding 3 — summary retrieval beats raw@7k retrieval (the cost question)

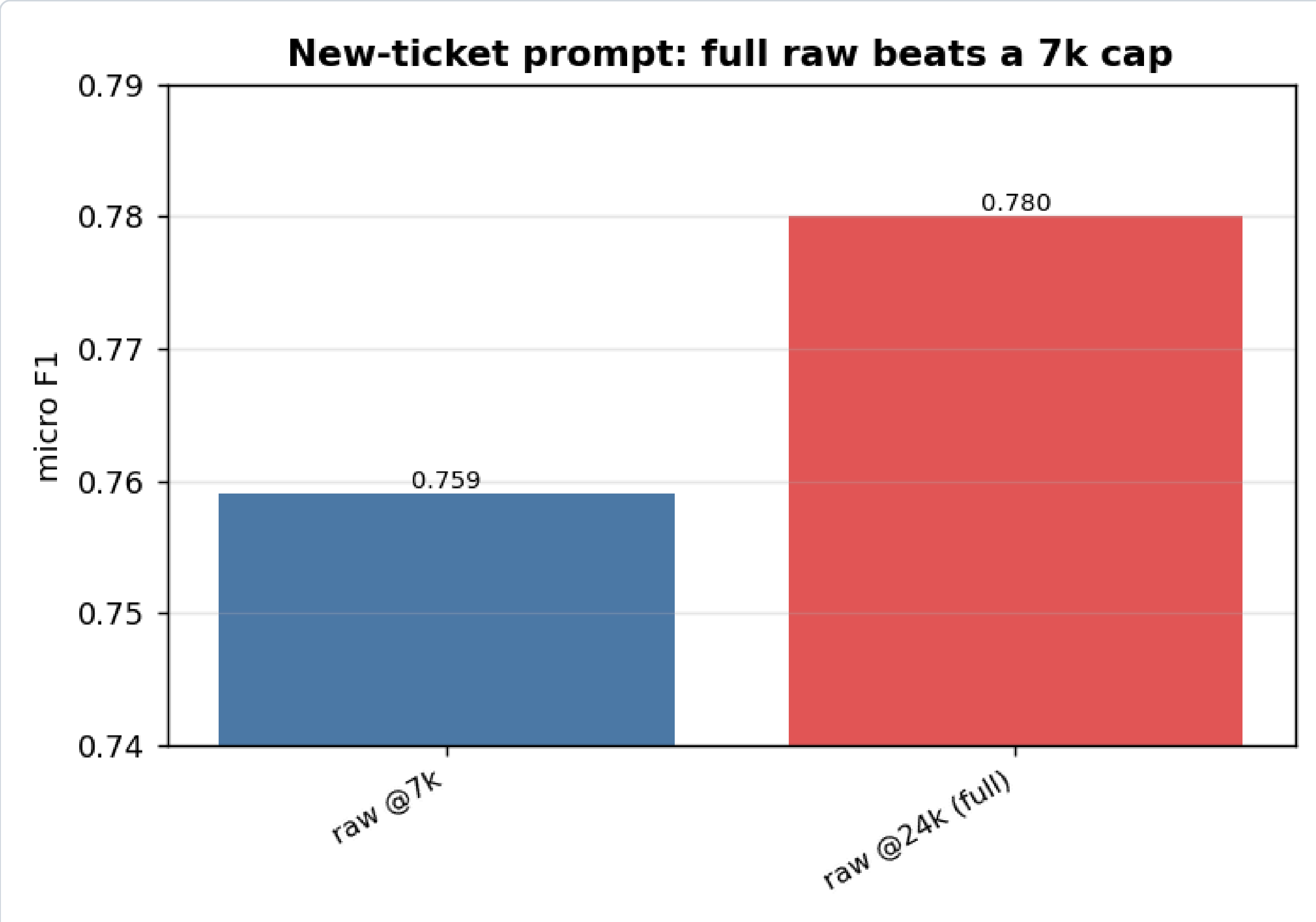


To test dropping summarization entirely, we re-embedded the index on **raw@7k** (no summary) and retrieved with raw. The decider is **historic-lane recall** — did the precedent search surface the GT (in evidence mode the 50-VS candidate pool makes review-pool recall structurally 1.0, so *that* isn't the signal — the historic lane is).

- summary retrieval: historic-lane R **0.902** → F1 **0.786**
- raw@7k retrieval: historic-lane R **0.843** → F1 **0.754**

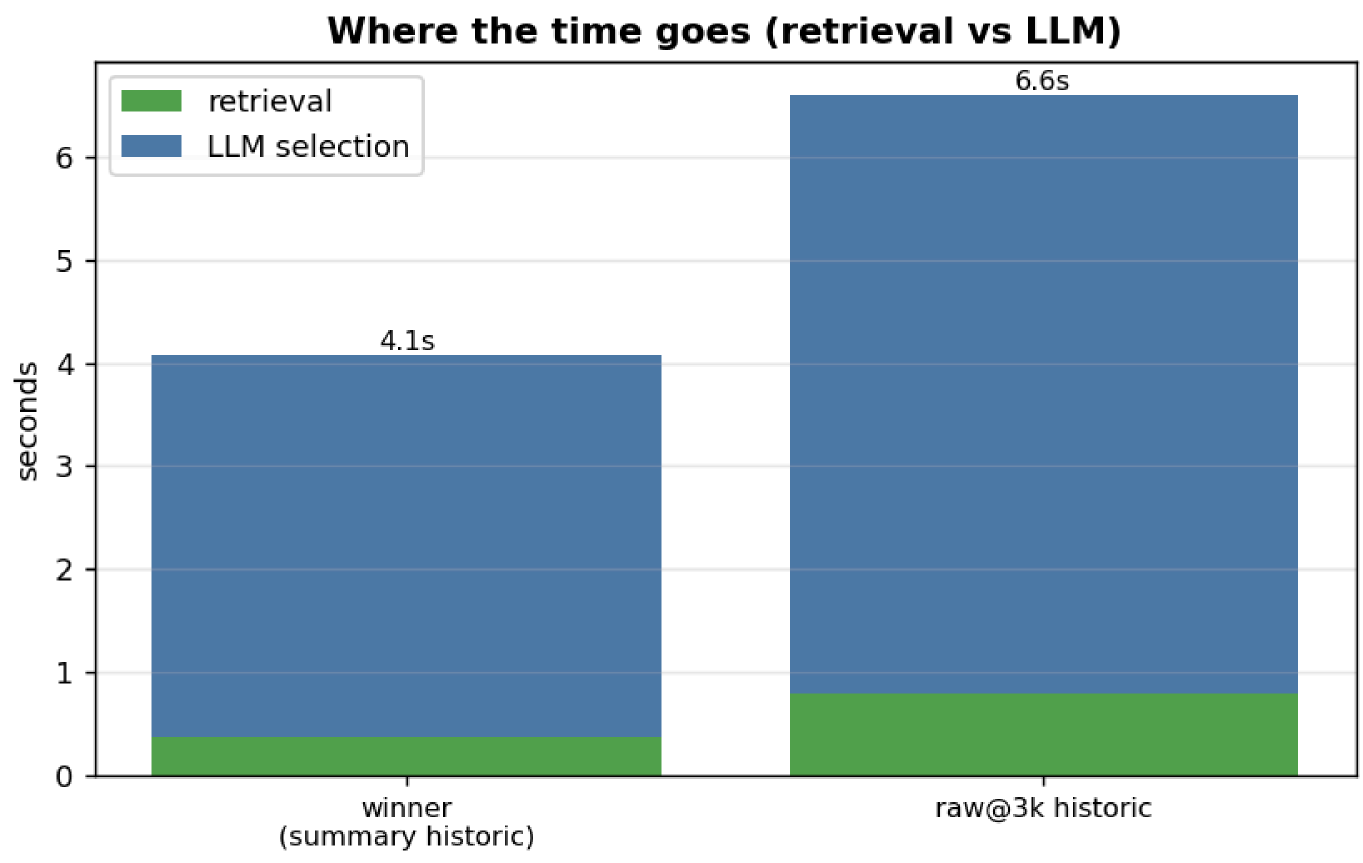
How to read it: the ~460-token summary embeds the *whole* ticket cleanly; raw@7k keeps only ~half of a big ticket, so it retrieves **worse neighbours**. Since precedent drives recall (GT *backed* by historic is picked ~0.82, *not-backed* only ~0.38), worse precedent → lower recall → ~3 F1 points lost. **Dropping the summary costs quality.**

Finding 4 — feed the new ticket's FULL raw; don't cap it



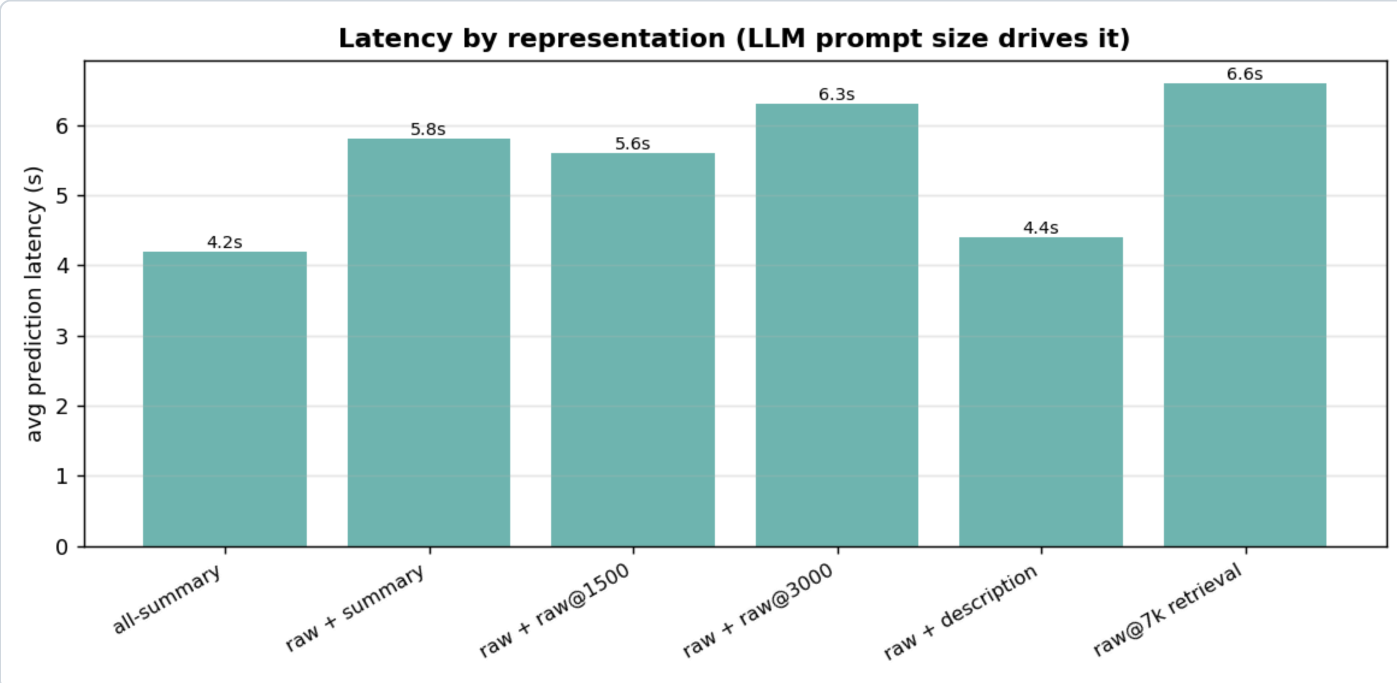
Holding everything else at the winner (summary retrieval + summary historic) and changing only the new ticket's raw cap: the **full ~24k raw** scores **0.780** vs **0.759** at a 7k cap — capping loses ~2 points and drops exact-set 31% → 26%. The extra context genuinely helps the LLM decide, and latency was identical (~4s) either way, so **there's no reason to cap the new-ticket prompt.**

Where the time goes (latency split)



How to read it: splitting prediction latency into retrieval vs the LLM call shows **retrieval is sub-second** (0.38–0.79s) — never the bottleneck. The **LLM call is the whole cost**, and it tracks the **historic block size**: summary historic 3.7s vs raw@3k historic 5.8s. So the cheap lever for latency is the historic representation (keep it summary), not the new-ticket prompt.

Latency / cost



How to read it: latency tracks the LLM **prompt size**. summary historic ≈ 6×460 tokens; raw@7k historic ≈ 6×7k = 42k tokens, which blows the prompt to ~55k tokens and spikes prediction to 132s on big-neighbour tickets. This is a *second* cost axis, independent of the summary question: even if you keep summaries, **never ship raw@7k historic** — it's ~4–5× the runtime token cost for zero quality.

Verdict

Locked config: summary retrieval + FULL ~24k raw new-ticket prompt + summary historic — F1 ≈0.78–0.79.

- The summary you "can't eliminate" turns out to be the thing buying your best retrieval.
- Dropping summarization (raw@7k retrieval) is a **regression** (0.780 → 0.742, historic-lane R 0.902 → 0.840), not a marginal trade. The summary-free idea is dead.
- The new-ticket **raw** prompt is the real win (+0.07) — and feed it **in full**; a 7k cap costs ~2pts.
- Use the **cheapest** historic block (summary): same quality as raw, and it's what keeps the LLM call (the only real latency) at ~3.7s. Retrieval is sub-second regardless.

Final config (locked) — the per-ticket flow

```
1. New ticket raw, consolidated to <=24k tokens
2. Summarize it (condense LLM)  -> summary fields
3. Embed the summary -> search the index -> retrieve the 6 nearest past tickets
```

```
4. Build the selection prompt:
      - the new ticket's FULL raw (~24k)                ("raw to decide")
      - the 6 neighbours' summaries + their VS labels     ("summaries as precedent")
      - the 50-VS catalogue
5. LLM picks the VS  (count = requested; evidence mode, Recall prompt)
```

The two uses of "summary" are different field sets, by design: - **Retrieval query** embeds **all six** summary fields (generatedSummary + businessProblem + businessCapability + keyTerms + stakeholders + systemsAndProducts). - **Historic block** shows each neighbour's **generatedSummary only** + its VS labels.

So every new ticket contributes **both** its summary (to *find* precedent) and its full raw (to *decide*) — which is exactly why summarization stays in the pipeline. **No further changes.**